

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Mechanical (CAD/CAM) Examination

January 2014

ME 704.01 Advanced Machine Design

Date: 01.01.2014, Wednesday Time: 10:00 a.m. To 01:00 p.m. Maximum Marks: 70

Instructions:

1. Attempt all the questions.
2. Section I and II must be attempted in separate answer book.
3. Make suitable assumptions and draw neat figures wherever if required.
4. Use of scientific calculator is allowed.
5. Figures to the right indicate the full marks.
6. PSG Design data book is allowed in examination.

SECTION - I

- Q.1 Answer the following. (any three)** [15]
- (a) Explain the constructional detail of a conveyor belt in brief with a neat sketch. [05]
- (b) Explain the important terms for conveyor design. [05]
- (i) Angle of repose and Angle of surcharge.
- (ii) Flowability
- (iii) Effective belt width for the material
- (iv) Volume capacity of belt
- (v) Mass capacity of belt
- (c) Explain the correction factor to avoid undercutting for corrective gear design. [05]
- (d) Explain basic principles in selection of material handling equipment. [05]
- Q.2** An inclined belt conveyor is used for transporting ash over a distance of 1.2 km and a height of 500 meters. The material density for ash can be taken as 0.5 t/m^3 . Assume the belt speed to be 1.5 m/s and belt width to be 400mm. Determine the capacity of the conveyor. For the belt, use the following data for the flowability factor. [08]

Conveyor Inclination	$10^\circ-15^\circ$	$16^\circ-20^\circ$	$21^\circ-25^\circ$	$26^\circ-30^\circ$	$31^\circ-35^\circ$
Flowability Factor, CI	2.65×10^{-4}	2.50×10^{-4}	2.35×10^{-4}	2.20×10^{-4}	2.05×10^{-4}

The effective width b (in meters), of the material carried by the belt safely is given by the following equation:

$$b = 0.9B - 0.05$$

where,

B = Belt width, m

OR

- Q.2** A horizontal belt conveyor is used for transporting a material of having mass density of 800 kg/m^3 . The surcharge factor for the belt drive is 0.08. Determine the belt width [08]

required, if the material is to be conveyed at a rate of 36 tonnes per hour. Assume the belt speed to be 1.75 m/s. The effective width b (in meters), of the material carried by the belt safely is given by the following equation,

$$b = 0.9B - 0.05$$

where,

B = Belt width, m

- Q.3 Answer the following. (any two)** [12]
- (a) Explain the characteristics of corrected gears. [06]
 - (b) Explain method of loading and discharging of conveyors. [06]
 - (c) Explain the classification of material handling equipments in brief. [06]

SECTION - II

- Q.4 Answer the following. (any three)** [15]
- (a) Explain the fracture in detail. [05]
 - (b) Explain different modes of fracture. [05]
 - (c) Explain stress intensity factor of crack in finite bodies. [05]
 - (d) Explain various quadrants of legal and ethical domain. [05]

- Q.5 Design a tensile bar of length $L=200\text{mm}$ to carry a tensile load of 5 kN for minimum cost of the following materials:** [08]

Material	Mass Density (kg/m^3)	Material Cost (Rs/N Weight)	Yield Strength (MPa)
A	7500	16	130
B	3000	32	50
C	4800	480	90
D	2100	32	20

OR

- Q.5 A machine component is subjected to a flexural stress which fluctuates between $+300 \text{ MN/m}^2$ and -150 MN/m^2 . Determine the value of minimum ultimate strength according to (i) Gerber Relation (ii) Modified Goodman relation (iii) Soderberg relation. Take yield strength = 0.55 ultimate strength, Endurance strength = 0.5 ultimate strength and FOS=2.** [08]

- Q.6 Answer the following. (any two)** [12]
- (a) Explain the importance of rigidity in design of machine tools and machines. [06]
 - (b) State the different theory of failure. Explain any two of them. [06]
 - (c) Discuss Human factor consideration in design and applications in brief. [06]
 - (d) Discuss the comparison between concurrent engineering and traditional engineering. [06]
